

```
const _ = require('lodash');  
let product = [  
  {name: "Laptop", category: "Elektronik", stock: 50, pricePerUnit: 15000000},  
  {name: "Smartphone", category: "Elektronik", stock: 100, pricePerUnit: 5000000}  
]
```

// Buat fungsi 'addProduct(inventory, product)' untuk menambahkan produk baru ke inventaris.

```
function addProduct(inventory, product){  
  inventory.push(product);  
  console.log(`Produk ${product.name} berhasil di tambahkan!`);  
  return inventory;  
}  
addProduct(product, {name: "Lipstik", category: "Kosmetik", stock: 100, pricePerUnit: 300000});
```

//Buat fungsi 'removeProduct(inventory, productName)' untuk menghapus produk dari inventaris berdasarkan nama produk.

```
function removeProduct(inventory, productName){  
  let index = -1;  
  for(i=0; i < inventory.length; i++){  
    if(inventory[i].name === productName){  
      index = i;  
      break;  
    }  
  }  
  inventory.splice(index, 1);  
  console.log(`Produk ${productName} berhasil dihapus!`)  
}  
  
removeProduct(product, "Smartphone");
```

//Buat fungsi 'updateProduct(inventory, productName, newDetails)' untuk memperbarui informasi produk tertentu.

```
function updateProduct(inventory, productName, newDetails){  
  let index = -1;  
  for(i=0; i < inventory.length; i++){  
    if(inventory[i].name === productName){  
      inventory[i].name = newDetails.name;  
      inventory[i].category = newDetails.category;  
      inventory[i].stock = newDetails.stock;  
      inventory[i].pricePerUnit = newDetails.pricePerUnit;  
      break;  
    }  
  }  
  console.log(`Produk ${productName} berhasil diperbaharui!`);  
}
```

```
updateProduct(product, "Lipstik", {name: "Lipstik", category: "Kosmetik", stock: 100, pricePerUnit:  
2000});
```

//Buat fungsi 'findMostExpensiveProduct(inventory)' menggunakan Lodash untuk menemukan produk dengan harga tertinggi.

```
function findMostExpensiveProduct(inventory){  
  const mostExpensive = _.maxBy(inventory, `pricePerUnit`);  
  return mostExpensive? mostExpensive : `Inventaris kosong.`;  
}  
  
const result = findMostExpensiveProduct(product);  
console.log("Produk termahal:", result);
```

// Buat fungsi 'calculateTotalInventoryValue(inventory)'

```
function calculateTotalInventoryValue(inventory) {  
    const totalValues = _.map(inventory, product => product.stock * product.pricePerUnit);  
    return _.sum(totalValues);  
}  
  
const totalValue = calculateTotalInventoryValue(product);  
console.log("Total nilai inventaris:", totalValue);
```

// Buat fungsi 'filterProductsByCategory(inventory, category)'

```
function filterProductsByCategory(inventory, category) {  
    return _.filter(inventory, { category });  
}  
  
const electronics = filterProductsByCategory(product, `Elektronik`);  
console.log(`Produk Elektronik: `, electronics);
```

HASIL RUNING CODE

```
PS D:\pangestu\pangestu nitip\Kumpulan assignment\assignment day 7> node day7.js  
● Produk Lipstik berhasil di tambahkan!  
Produk Smartphone berhasil dihapus!  
Produk Lipstik berhasil diperbaharui!
```

Ketika di console.log:

```
[  
  {  
    name: 'Laptop',  
    category: 'Elektronik',  
    stock: 50,  
    pricePerUnit: 15000000  
  },  
  {  
    name: 'Lipstik',  
    category: 'Kosmetik',  
    stock: 100,  
    pricePerUnit: 2000  
  }  
]
```

Analisi data inventaris

```
PS D:\pangestu\pangestu nitip\Kumpulan assignment\assignment day 7> node analisis.js
● Produk termahal: {
  name: 'Laptop',
  category: 'Elektronik',
  stock: 50,
  pricePerUnit: 15000000
}
Total nilai inventaris: 1250000000
Produk Elektronik: [
  {
    name: 'Laptop',
    category: 'Elektronik',
    stock: 50,
    pricePerUnit: 15000000
  },
  {
    name: 'Smartphone',
    category: 'Elektronik',
    stock: 100,
    pricePerUnit: 5000000
  }
]
```